THE ASPECT RATIO OF THE TWIN DRAGON IS $1/\varphi$

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Abstract. We show that the geometric aspect ratio of the Twin Dragon — defined via the Gaussian integer $1+i$ — equals $1/\varphi$, where $\varphi = (1+\sqrt{5})/2$ is the golden ratio, and that its major axis makes angle $-\arctan\varphi$ with the horizontal. More generally, for every equal-contraction IFS $\{f_k(z) = a(s_k z + t_k)\}$ with rotations $s_k \in S^1$ (or reflections) and centred translations, we give closed-form formulas for the aspect ratio, ellipticity, and principal-axis angle of the attractor. When all rotations are trivial, the aspect ratio depends only on a and a single anisotropy parameter $\kappa \in [0,1]$; for $a = 1/\lambda$ with λ in an imaginary quadratic ring, it lies in the quadratic field $\mathbb{Q}(\sqrt{d}\,)$ where d is the square-free part of $\mathbf{N}(\lambda^2-1)$. Since $1/\varphi$ is the reciprocal of the first metallic mean, it is natural to ask which others arise: every metallic ratio $\mu_m = (m+\sqrt{m^2+4})/2$ ($m \geq 1$) arises as the reciprocal aspect ratio of a plane-filling tile over $\mathbb{Z}[i]$; moreover, $\mathbb{Z}[i]$ is the unique imaginary quadratic ring where collinear digits can produce metallic-ratio aspect ratios.

1. Introduction

The Twin Dragon is a classical plane-filling fractal [5] governed by the Gaussian integer $1+i$: its two defining contractions share the linear part $(1-i)/2$, and their translations are symmetric about the origin. Despite this simple, purely arithmetic construction, the attractor has a non-trivial shape that is not immediately obvious from the definition. Visually the Twin Dragon looks somewhat elongated — neither a square nor a thin sliver — which raises the question of its precise aspect ratio.

A natural quantitative measure of shape for a planar set is the *geometric aspect ratio*: the ratio of standard deviations along the principal axes of its area distribution, or equivalently the square root of the ratio of the eigenvalues of the second-moment (covariance) matrix M of the uniform measure on the attractor. This definition gives the expected answers for elementary shapes: an ellipse with semi-axes $\alpha \geq \beta > 0$ has aspect ratio β/α (eigenvalues $\alpha^2/4$ and $\beta^2/4$, so the definition directly recovers the ratio of the semi-axes); a $p \times q$ rectangle has aspect ratio p/q (eigenvalues $p^2/12$ and $q^2/12$); and a square has aspect ratio 1. The covariance matrix is the unique second-order shape descriptor of a probability distribution: any similarity-invariant quantity derived from second moments alone is a function of the eigenvalue ratio of M. Other notions of aspect ratio — the bounding-box ratio, the Feret diameter ratio, the eccentricity of the convex hull — require knowledge of the boundary or its extreme points, which for general IFS is not available in closed form. The covariance definition, by contrast, is determined entirely by the

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invariant measure and is exactly computable for self-similar IFS via the moment fixed-point equation.

The approach of characterising IFS attractors through moments of their invariant measures was introduced by Vrscay and Roehrig [13] and systematically developed in [14]. Further analysis of self-similar measures via moments and Fourier transforms appears in Strichartz [12] and Lau–Ngai [10]; however, explicit aspect-ratio formulas were not derived in those works. The key observation is that the self-similar measure satisfies a linear fixed-point equation in its moments, which can be solved exactly. When the open set condition holds, Hutchinson's theorem [7] guarantees that the self-similar measure coincides with the normalised Hausdorff measure $\mathcal{H}^s|_A$, so the covariance matrix captures the actual geometry of the attractor at whatever dimension s it carries. In the special case where the similarity dimension s equals the ambient dimension, $\mathcal{H}^s$ reduces to Lebesgue area.

We apply this method to the Twin Dragon. The covariance matrix turns out to be $M = \frac{1}{5}\begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix}$, whose eigenvalues involve $\sqrt{5}$, leading to the aspect ratio $1/\varphi$, where $\varphi = (1+\sqrt{5})/2$ is the golden ratio. The appearance of φ in a fractal with no pentagonal geometry is unexpected. Although the method of moments for IFS goes back to [13], the closed-form aspect ratio and orientation formulas (Theorem 2 below) and the characterisation of its quadratic field for Gaussian-integer parameters (Corollary 4) are, to our knowledge, new.

Discovery. The result was first found empirically using the IFStile software [11], which implements the moment invariants of [13, 14] as part of its algebraic IFS search engine. Among the similarity invariants reported by IFStile for the Twin Dragon, the aspect ratio $1/\varphi$ appeared as a numerical value; the present paper supplies the exact proof and its generalisation to arbitrary equal-contraction IFS with centred translations. This is an instance of *experimental mathematics* [4]: a computer experiment suggests a precise conjecture, which is then confirmed by a rigorous proof.

The paper is organised as follows. Section 2 derives closed-form formulas for the aspect ratio and principal-axis angle (Theorem 2) and classifies the quadratic field of the aspect ratio (Corollary 4). Section 3 applies the formula to a two-parameter family of plane-filling tiles parameterised by $x^2-\ell x+N$ (Proposition 10). Section 4 specialises to tiles over $\mathbb{Z}[i]$, proves that every metallic ratio μ_m arises as the reciprocal aspect ratio of such a tile (Theorem 14), and shows that $\mathbb{Z}[i]$ is the unique imaginary quadratic ring where collinear tiles achieve metallic-ratio aspect ratios (Corollary 19). Section 5 applies the formula to the Twin Dragon and proves the main result.

2. The aspect ratio formula

An *iterated function system* (IFS) is a finite collection of contractions $f_1, \dots, f_N$ on a complete metric space. By Hutchinson's theorem [7] there exists a unique non-empty compact set A, the *attractor*, satisfying $A = f_1(A) \cup \cdots \cup f_N(A)$. For background see [2, 6]; for self-affine digit tiles in particular, see [1, 8, 9].

Consider an equal-weight IFS of $N \geq 2$ contractions on $\mathbb{C}$, each of the form

$$f_k(z) = a(s_k z + t_k) \qquad \text{or} \qquad f_k(z) = a(s_k \overline{z} + t_k),$$

where $a \in \mathbb{C}$, $|a| < 1$, the rotations satisfy $s_k \in S^1$, and $t_1, \ldots, t_N \in \mathbb{C}$ satisfy $\sum_{k=1}^N t_k = 0$.

Remark 1 (Centering). The condition $\sum t_k = 0$ is no loss of generality. The translation $z \mapsto z - c$ replaces each t_k by $\tilde{t}_k = s_k c + t_k - c/a$ (preserving maps) or $\tilde{t}_k = s_k \bar{c} + t_k - c/a$ (reversing maps). Setting $\sum \tilde{t}_k = 0$ gives a linear equation in c (and $\bar{c}$ when reversing maps are present) whose unique solvability is guaranteed by $|a| < 1$. When all s_k coincide, this simplifies to $t_k \mapsto t_k - \bar{t}$, a uniform shift of the digits.

Define the *digit variance* $\sigma^2 := \frac{1}{N}\sum_k |t_k|^2 > 0$, the *algebraic variance* $\tau := \frac{1}{N}\sum_k t_k^2 \in \mathbb{C}$, and the *digit anisotropy*

$$\kappa := \frac{|\tau|}{\sigma^2} = \frac{|\sum t_k^2|}{\sum |t_k|^2} \in [0,1].$$

The bound $\kappa \le 1$ follows from the triangle inequality $|\sum t_k^2| \le \sum |t_k^2| = \sum |t_k|^2$. When all t_k are collinear ($t_k = e^{i\theta} r_k$ with $r_k \in \mathbb{R}$), one has $\kappa = 1$; vertices of a regular N-gon ($N \ge 3$) give $\kappa = 0$. The Twin Dragon is the case $N = 2$, $a = (1-i)/2$, $s_k = 1$, $t_k = \pm 1$ ($\sigma^2 = 1$, $\kappa = 1$), studied in Section 5.

Theorem 2. *With the notation above, let I and J denote the sets of orientation-preserving and -reversing indices, and define*

$$u = \frac{1}{N}\sum_{k\in I} s_k^2, \qquad v = \frac{1}{N}\sum_{k\in J} s_k^2.$$

Then the second moment $\omega := E_\mu[Z^2]$ is

$$\omega = \frac{a^2\big(\tau + \overline{a^2}(v\,\overline{\tau} - \bar{u}\,\tau)\big)}{|1 - a^2 u|^2 - |a|^4 |v|^2}, \tag{1}$$

the aspect ratio is $\mathrm{AR}^2 = (1-\rho)/(1+\rho)$, where

$$\rho = \frac{(1-|a|^2)\,|\omega|}{|a|^2\,\sigma^2}, \tag{2}$$

and the major axis of the attractor makes angle

$$\psi = \tfrac{1}{2}\arg\omega \tag{3}$$

with the positive x-axis. When $v = 0$ (all maps orientation-preserving), (1) reduces to $\omega = a^2\tau/(1 - a^2 u)$, and

$$\rho = \frac{(1-|a|^2)\,|\tau|}{|1 - a^2 u|\,\sigma^2}. \tag{4}$$

In general, the aspect ratio depends on u, v, and the complex value of τ (not just $|\tau|$).

Proof. Let Z be distributed according to the invariant measure μ. The fixed-point equation

$$E[h(Z)] = \frac{1}{N}\Big(\sum_{k\in I} E\big[h\big(a(s_k Z + t_k)\big)\big] + \sum_{k\in J} E\big[h\big(a(s_k \overline{Z} + t_k)\big)\big]\Big) \tag{5}$$

applied to $h(z) = z$ gives $E[Z] = a\big(u'\,E[Z] + v'\,\overline{E[Z]} + \frac{1}{N}\sum t_k\big)$ where $u' = \frac{1}{N}\sum_{k\in I} s_k$, $v' = \frac{1}{N}\sum_{k\in J} s_k$; since $|a|(|u'| + |v'|) \le |a| < 1$ and $\sum t_k = 0$, the unique solution is $E[Z] = 0$.

Applying (5) to $h(z) = |z|^2$: since $E[Z] = 0$, the cross terms $2\,\mathrm{Re}(\overline{s_k Z}\, t_k)$ vanish in expectation, and $|s_k z|^2 = |z|^2$, giving

$$E[|Z|^2] = |a|^2\big(E[|Z|^2] + \sigma^2\big),$$

so $E[|Z|^2] = |a|^2\sigma^2/(1 - |a|^2)$.

Applying (5) to $h(z) = z^2$ and writing $\omega := E[Z^2]$: $(s_k z)^2 = s_k^2 z^2$ for preserving maps and $s_k^2 \overline{z}^2$ for reversing ones, while the cross terms $2s_k z{\cdot}t_k$ vanish in expectation and the translations contribute $\sum t_k^2 = N\tau$, so

$$\omega = a^2\big(u\,\omega + v\,\overline{\omega} + \tau\big). \tag{6}$$

Rewriting as the $\mathbb{R}$-linear system $(1 - a^2 u)\,\omega - a^2 v\,\overline{\omega} = a^2\tau$ and applying Cramer's rule ($\Delta := |1 - a^2 u|^2 - |a|^4|v|^2 > 0$ since $|a|^2(|u| + |v|) \le |a|^2 < 1$) gives (1). The ellipticity $\rho = |\omega|/E[|Z|^2]$ gives (2), and $\mathrm{AR}^2 = (1-\rho)/(1+\rho)$. The orientation (3) follows from the covariance decomposition in Remark 9.

When $v = 0$, the equation is $\mathbb{C}$-linear: $\omega = a^2\tau/(1 - a^2 u)$, giving (4). □

When all rotations are trivial, the formulas simplify to depend on a single anisotropy parameter.

Corollary 3. *When all $s_k = 1$ (so $u = 1$, $v = 0$), the aspect ratio depends on the translations only through $\kappa = |\tau|/\sigma^2 \in [0, 1]$:*

$$\mathrm{AR}^2 = \frac{|1 - a^2| - (1 - |a|^2)\,\kappa}{|1 - a^2| + (1 - |a|^2)\,\kappa}. \tag{7}$$

(a) *For fixed a, AR is strictly decreasing in κ: $\kappa = 0$ (isotropic digit set) gives $\mathrm{AR} = 1$, and $\kappa = 1$ (collinear translations) gives the most elongated attractor.*
(b) *$\mathrm{AR} = 0$ if and only if $a \in \mathbb{R}$ and $\kappa = 1$.*

For collinear translations ($\kappa = 1$, $\tau > 0$ after aligning the digit axis with $\mathbb{R}$),

$$\tan 2\psi = \frac{\mathrm{Im}(a^2)}{\mathrm{Re}(a^2) - |a|^4}\,; \tag{8}$$

for non-collinear digits, ψ is shifted by $\frac{1}{2}\arg\tau$.

Proof. Setting $s_k = 1$ gives $u = 1$, $v = 0$, and (4) yields $\rho = (1 - |a|^2)\kappa/|1 - a^2|$, giving (7). Properties (a)–(b) follow: ρ is increasing in κ, vanishes at $\kappa = 0$, and $\rho = 1$ (i.e. $\mathrm{AR} = 0$) requires $\kappa = 1$ and $|1 - a^2| = 1 - |a|^2$, which holds iff $a \in \mathbb{R}$. □

The aspect ratio in (7) involves $|1 - a^2|$, which is generically transcendental. When $a = 1/\lambda$ for an algebraic integer λ in an imaginary quadratic field $K = \mathbb{Q}(\sqrt{-D})$, this modulus reduces to $\sqrt{\mathbf{N}(\lambda^2 - 1)}/\mathbf{N}(\lambda)$ (where $\mathbf{N}$ is the field norm, $\mathbf{N}(\alpha) = |\alpha|^2 \in \mathbb{Z}$ for $\alpha \in \mathcal{O}_K$), and the aspect ratio becomes a quadratic irrational.

Corollary 4. *Let $K = \mathbb{Q}(\sqrt{-D})$ be an imaginary quadratic field with ring of integers $\mathcal{O}_K$, let $\lambda \in \mathcal{O}_K$ with $|\lambda| > 1$ and $\lambda \notin \mathbb{R}$, and set $a = 1/\lambda$. For any IFS as in Corollary 3 with $\kappa = 1$, the aspect ratio (7) lies in $\mathbb{Q}(\sqrt{d}\,)$, where d is the square-free part of $\mathbf{N}(\lambda^2 - 1)$. Explicitly,*

$$\mathrm{AR}^2 = \frac{\sqrt{\mathbf{N}(\lambda^2 - 1)} - (\mathbf{N}(\lambda) - 1)}{\sqrt{\mathbf{N}(\lambda^2 - 1)} + (\mathbf{N}(\lambda) - 1)}. \tag{9}$$

Proof. For $\lambda \in \mathcal{O}_K$, $\mathbf{N}(\lambda) = |\lambda|^2 \in \mathbb{Z}$ and $\mathbf{N}(\lambda^2 - 1) = |\lambda^2 - 1|^2 \in \mathbb{Z}$. Substitute $|a|^2 = 1/\mathbf{N}(\lambda)$, $|1 - a^2| = \sqrt{\mathbf{N}(\lambda^2 - 1)}/\mathbf{N}(\lambda)$, and $\kappa = 1$ into (7). □

TABLE 1. Aspect ratios for small Gaussian integers $\lambda \in \mathbb{Z}[i]$, illustrating Corollary 4. The quadratic field $\mathbb{Q}(\sqrt{d}\,)$ is determined by the square-free part of $\mathbf{N}(\lambda^2 - 1)$.

λ	$\lambda^2 - 1$	$\mathbf{N}(\lambda^2 - 1)$	d	AR
$1+i$	$-1+2i$	5	5	$1/\varphi = (\sqrt{5}-1)/2 \approx 0.618$
$2+i$	$2+4i$	20	5	$1/\varphi^3 = \sqrt{5}-2 \approx 0.236$
$1+2i$	$-4+4i$	32	2	$\sqrt{2}-1 \approx 0.414$

The first two rows of Table 1 share the same quadratic field $\mathbb{Q}(\sqrt{5})$ because the rational prime 5 divides both norms. This is not a coincidence: the Cayley-type transform (9) maps $\sqrt{\mathbf{N}(\lambda^2-1)}$ to an element of the real quadratic field $\mathbb{Q}(\sqrt{d}\,)$, and the particular metallic ratio that appears is controlled by the arithmetic of this field. In the first row, $\sqrt{5}$ yields a rational expression in φ thanks to the identity $\varphi^2 = \varphi + 1$: the golden ratio is the fundamental unit of the ring $\mathbb{Z}[\varphi]$, so every element of $\mathbb{Q}(\sqrt{5})$ in $(0,1)$ of the form $(\sqrt{5}-n)/(\sqrt{5}+n)$ is a power of $1/\varphi$. The third row gives the *silver ratio* $\delta_S = \sqrt{2}-1$ (fundamental unit of $\mathbb{Z}[\sqrt{2}]$).

All examples in Table 1 have $s_k = 1$. When rotations are present, the general formula (4) applies, and the aspect ratio can change dramatically.

Example 5 (Lévy C curve)**.** Take the same contraction $a = (1-i)/2$ and centred digits $t_0 = -1$, $t_1 = 1$ as in the Twin Dragon, but set $s_0 = i$, $s_1 = 1$ (so f_0 includes an extra 90° rotation). The attractor is a copy of the *Lévy C curve*. Since $\sigma^2 = 1$, $\tau = 1$, $\kappa = 1$, the digit geometry is identical to the Twin Dragon; only the rotation average changes: $u = (i^2 + 1^2)/2 = 0$. By (4),

$$\rho = \frac{(1-|a|^2)\,|\tau|}{|1-a^2u|\,\sigma^2} = \frac{\frac{1}{2}\cdot 1}{|1-0|\cdot 1} = \frac{1}{2},$$

whence $\mathrm{AR}^2 = (1-\frac{1}{2})/(1+\frac{1}{2}) = 1/3$, i.e. $\mathrm{AR} = 1/\sqrt{3} \approx 0.577$. For comparison, the Twin Dragon has $u = 1$ and $\mathrm{AR} = 1/\varphi \approx 0.618$ (Table 1). Thus a single rotation $s_0 \mapsto i$ replaces $u = 1$ with $u = 0$, changing the aspect ratio from $1/\varphi$ to $1/\sqrt{3}$. More generally, for $N = 2$ one has $u = 0$ if and only if $s_1/s_0 = \pm i$ (the two rotations differ by 90°). The *Heighway Dragon*, with $s_0 = -1$ and $s_1 = i$, is another instance: its attractor has a very different shape but the same $\mathrm{AR}^2 = 1/3$.

Example 6 (Conjugate parallelogram)**.** Replace the two preserving maps of the Twin Dragon by their orientation-reversing counterparts: $f_k(z) = a(\overline{z} \pm 1)$, $a = (1-i)/2$. The digits $t_k = \pm 1$ and contraction a are unchanged, so $\sigma^2 = 1$, $\tau = 1$, $\kappa = 1$; but now $I = \emptyset$, $J = \{1,2\}$, giving $u = 0$, $v = 1$. Setting $u = 0$, $v = 1$, $\tau = 1$ in (1) gives

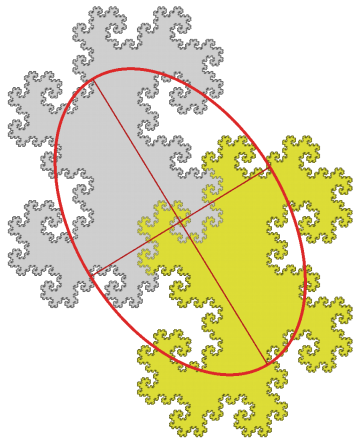
$$\omega = \frac{a^2(1+\overline{a^2})}{1-|a|^4} = \frac{(-i/2)(1+i/2)}{3/4} = \frac{1-2i}{3}\,,$$

so $|\omega| = \sqrt{5}/3$ and $\rho = \sqrt{5}/3$. Therefore

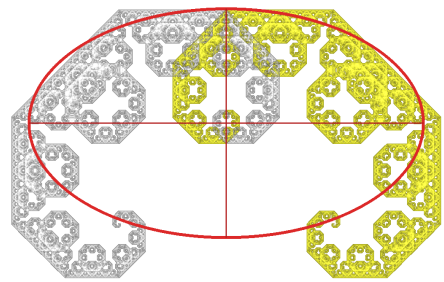
$$\mathrm{AR}^2 = \frac{3-\sqrt{5}}{3+\sqrt{5}} = \frac{7-3\sqrt{5}}{2} = \frac{1}{\varphi^4}\,, \qquad \mathrm{AR} = \frac{1}{\varphi^2} = \frac{3-\sqrt{5}}{2} \approx 0.382.$$

The attractor is a parallelogram with vertices $\pm i$ and $\pm(2-i)$ (one verifies that both maps permute these four vertices). It is more elongated than the Twin Dragon ($1/\varphi^2 < 1/\varphi$). Thus replacing z by $\overline{z}$ in both maps (switching from $u = 1$, $v = 0$

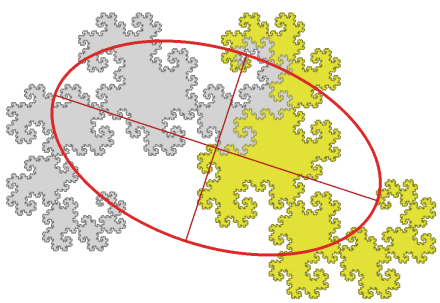
to $u = 0$, $v = 1$) changes the aspect ratio from $1/\varphi$ to $1/\varphi^2$, while keeping it in $\mathbb{Q}(\sqrt{5})$.



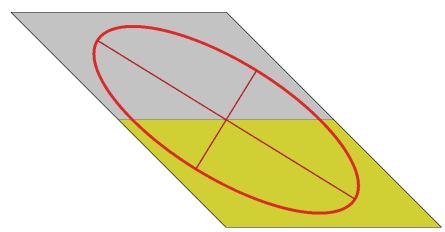
(a) Twin Dragon



(b) Lévy C curve



(c) Heighway Dragon



(d) Conjugate parallelogram

FIGURE 1. Four IFS with contraction $a = (1 - i)/2$ and centred digits in $\mathbb{Z}[i]$, shown with their 1.5σ covariance ellipses. The Twin Dragon (a) has $s_k = 1$ and $\mathrm{AR} = 1/\varphi$. The Lévy C curve (b) and Heighway Dragon (c) both have $u = 0$ and $\mathrm{AR} = 1/\sqrt{3}$: the ellipses share the same aspect ratio but have different orientations ($\psi = 0°$ and $\psi \approx 18°$, respectively). The Conjugate parallelogram (d) has all maps orientation-reversing ($u = 0$, $v = 1$) and $\mathrm{AR} = 1/\varphi^2$.

Corollary 4 assumes $\kappa = 1$. The following extension covers arbitrary κ.

Corollary 7. *For any IFS $\{f_k(z) = a(z + t_k)\}$ with $a = 1/\lambda$, $\lambda \in \mathcal{O}_K \setminus \mathbb{R}$ (K an imaginary quadratic field), and digits $t_k \in \mathcal{O}_K$, the anisotropy satisfies $\kappa^2 \in \mathbb{Q}$, and AR^2 lies in $\mathbb{Q}(\sqrt{d}\,)$ where d is the square-free part of $\mathbf{N}(\lambda^2 - 1) \cdot \big(\mathbf{N}(\lambda^2 - 1) - \kappa^2(\mathbf{N}(\lambda) - 1)^2\big)$. In particular, the quadratic field of AR depends only on λ and κ^2, not on the individual digits.*

Proof. Write $\tau = \sum (t_k - \bar{t})^2 \in \mathcal{O}_K$ and $\sigma^2 = \sum |t_k - \bar{t}|^2 \in \mathbb{Z}$. Then $\kappa^2 = |\tau|^2/\sigma^4 = \mathbf{N}(\tau)/\sigma^4$, which is rational since $\mathbf{N}(\tau) \in \mathbb{Z}$. Substituting into (7) gives

$$\mathrm{AR}^2 = \frac{|1-a^2| - (1-|a|^2)\kappa}{|1-a^2| + (1-|a|^2)\kappa} = \frac{\sqrt{R} - (N-1)\kappa}{\sqrt{R} + (N-1)\kappa},$$

where $R = \mathbf{N}(\lambda^2 - 1)/N^2$. Clearing N^2: the numerator and denominator are $\sqrt{\mathbf{N}(\lambda^2-1)} \mp (N-1)\kappa$. Rationalising:

$$\mathrm{AR}^2 = \frac{\sqrt{\mathbf{N}(\lambda^2-1)} - (N-1)\kappa}{\sqrt{\mathbf{N}(\lambda^2-1)} + (N-1)\kappa} = \frac{\mathbf{N}(\lambda^2-1) - (N-1)^2\kappa^2}{\left(\sqrt{\mathbf{N}(\lambda^2-1)} + (N-1)\kappa\right)^2}.$$

Now $\mathbf{N}(\lambda^2-1) \in \mathbb{Z}$ and $(N-1)^2\kappa^2 = (N-1)^2\mathbf{N}(\tau)/\sigma^4 \in \mathbb{Q}$, so AR^2 is a rational function of $\sqrt{\mathbf{N}(\lambda^2-1)}$ and lies in $\mathbb{Q}(\sqrt{d}\,)$. □

Remark 8. For collinear translations ($\kappa = 1$), formula (7) shows that AR depends on a only through $(|a|^2, |1-a^2|)$: the attractor is nearly round when $|1-a^2|$ is large relative to $1-|a|^2$, and degenerates to a segment when $|1-a^2| \approx 1-|a|^2$ (i.e. a nearly real). For every a in the family, $\mathrm{AR}(a) \le |a|$, with equality if and only if a is purely imaginary (the attractor is then a rectangle, cf. §3). Indeed, $|1-a^2| \le 1+|a|^2$ (triangle inequality), with equality iff $a^2/|a^2| = -1$, i.e. $a \in i\mathbb{R}$. Substituting $|1-a^2| = 1+|a|^2$ into (7) gives $\mathrm{AR}^2 = |a|^2$. Figure 2 shows AR over the entire sub-family $|a| = 1/\sqrt{2}$.

Remark 9. The orientation formula (3) follows from the decomposition of the covariance matrix. Writing $\omega = |\omega|\, e^{2i\psi}$ for $Z = X + iY$ with $E[Z] = 0$:

$$M = \frac{E[|Z|^2]}{2}\, I + \frac{|\omega|}{2} \begin{pmatrix} \cos 2\psi & \sin 2\psi \\ \sin 2\psi & -\cos 2\psi \end{pmatrix},$$

whose larger eigenvalue has eigenvector $(\cos\psi, \sin\psi)$. For the collinear case (8), $\mathrm{Im}(a^2)$ measures the rotation per level of the IFS hierarchy, and $\mathrm{Re}(a^2) - |a|^4$ is the real drift; their ratio determines the principal-axis direction.

3. The $\lambda_{N,\ell}$ family

A natural two-parameter family of plane-filling tiles arises by applying Corollary 3 to collinear digits. For integers $N \ge 2$ and $0 \le \ell < 2\sqrt{N}$, let $\lambda_{N,\ell}$ be a root of $x^2 - \ell x + N$ (so $|\lambda_{N,\ell}|^2 = N$ and $\lambda_{N,\ell} \notin \mathbb{R}$). The digits $\{0, 1, \ldots, N-1\}$ form a complete residue system (CRS) of $\mathbb{Z}[\lambda_{N,\ell}]$ modulo $\lambda_{N,\ell}$ [8, 9]. Indeed, every element $a + b\lambda_{N,\ell} \in \mathbb{Z}[\lambda_{N,\ell}]$ satisfies $\lambda_{N,\ell}(a + b\lambda_{N,\ell}) = a\lambda_{N,\ell} + b(\ell\lambda_{N,\ell} - N)$ (using $\lambda_{N,\ell}^2 = \ell\lambda_{N,\ell} - N$), whose integer part is $-bN$; hence $a + b\lambda_{N,\ell} \equiv 0 \pmod{\lambda_{N,\ell}}$ forces $b = 0$ and $N \mid a$, so $\lambda_{N,\ell}\mathbb{Z}[\lambda_{N,\ell}] \cap \mathbb{Z} = N\mathbb{Z}$ and the N residues $\{0, \ldots, N-1\}$ are distinct. The N-map IFS with $a = 1/\lambda_{N,\ell}$ and digits centred to $\sum t_j = 0$ therefore produces a tile. The aspect ratio and orientation of these tiles admit explicit formulas.

Proposition 10. *Let $N \ge 2$ and $0 \le \ell < 2\sqrt{N}$, and let $\lambda = \lambda_{N,\ell}$ be a root of $x^2 - \ell x + N$. For the N-map tiling IFS with contraction $a = 1/\lambda$:*

(i) *$|\lambda^2 - 1|^2 = (N+1)^2 - \ell^2$, and the aspect ratio is*

$$\mathrm{AR}_{N,\ell} = \sqrt{\frac{\sqrt{(N+1)^2 - \ell^2} - (N-1)}{\sqrt{(N+1)^2 - \ell^2} + (N-1)}}. \tag{10}$$

(ii) *The major axis of inertia makes angle*

$$\psi_{N,\ell} = -\tfrac{1}{2}\arg(\lambda^2 - 1) \tag{11}$$

with the positive x-axis, where $\lambda^2 - 1 = \frac{1}{2}\big((\ell^2 - 2N - 2) + i\ell\sqrt{4N - \ell^2}\big)$.

(iii) *When* $\ell = 2$ *(the sub-family* $\lambda_N = 1 + i\sqrt{N-1}$*), (10) simplifies to* $\mathrm{AR}_N = (\sqrt{N+3} - \sqrt{N-1})/2$*, and* $\tan\psi$ *satisfies* $t^2 + t\sqrt{N-1} - 1 = 0$*. When additionally* $N - 1 = m^2$*, this is the m-th metallic-ratio equation:* $m = 1$ *gives* φ*,* $m = 2$ *gives* $1 + \sqrt{2}$*.*

Proof. Part (i): Expanding $\lambda = (\ell + i\sqrt{4N - \ell^2})/2$ gives

$$|\lambda^2 - 1|^2 = \tfrac{1}{4}\big[(\ell^2 - 2N - 2)^2 + \ell^2(4N - \ell^2)\big] = (N+1)^2 - \ell^2.$$

Substituting $|a|^2 = 1/N$ and $|1 - a^2| = \sqrt{(N+1)^2 - \ell^2}/N$ into (7) yields (10). Part (ii): for collinear digits $\omega = a^2\sigma^2/(1 - a^2) = \sigma^2/(\lambda^2 - 1)$, so (3) gives $\psi = -\frac{1}{2}\arg(\lambda^2 - 1)$. Part (iii): Setting $\ell = 2$ gives $(N+1)^2 - 4 = (N-1)(N+3)$; the AR formula rationalises to $(\sqrt{N+3} - \sqrt{N-1})/2$. For the tangent, $\lambda^2 - 1 = (1 - N) + 2i\sqrt{N-1}$; the double-angle identity $\tan 2\psi = 2\tan\psi/(1 - \tan^2\psi) = 2/\sqrt{N-1}$ yields the stated equation. □

Selected values:

N	ℓ	$\lambda_{N,\ell}$	field	$\mathrm{AR}_{N,\ell}$
2	0	$i\sqrt{2}$	$\mathbb{Q}$	$1/\sqrt{2} \approx 0.707$ (rectangle)
2	1	$(1 + i\sqrt{7})/2$	$\mathbb{Q}(\sqrt{2})$	≈ 0.691 (tame twindragon)
2	2	$1 + i$	$\mathbb{Q}(\sqrt{5})$	$1/\varphi \approx 0.618$
5	2	$1 + 2i$	$\mathbb{Q}(\sqrt{2})$	$\sqrt{2} - 1 \approx 0.414$
5	4	$2 + i$	$\mathbb{Q}(\sqrt{5})$	$1/\varphi^3 \approx 0.236$

The "field" column records $\mathbb{Q}(\sqrt{d}\,)$ where d is the square-free part of $(N+1)^2 - \ell^2$. The entry $(N, \ell) = (2, 1)$ recovers the tame twindragon (see below), confirming that it belongs to the same two-parameter family. The entry $(N, \ell) = (5, 4)$ gives $\lambda_{5,4} = 2 + i$ — the same contraction as the 2-map tile in row 2 of Table 1, illustrating again that AR depends only on $a = 1/\lambda$ (Corollary 3).

For fixed ℓ, $\mathrm{AR}_{N,\ell} \to 0$ as $N \to \infty$ (stronger contraction elongates the attractor). For fixed N, increasing ℓ toward $2\sqrt{N}$ drives a toward the real axis and $\mathrm{AR} \to 0$; setting $\ell = 0$ gives $\lambda = i\sqrt{N}$ (purely imaginary a) and $\mathrm{AR} = 1/\sqrt{N}$, the maximum for a given N. For $N = 2$ the IFS $\{(i/\sqrt{2})(z \pm 1)\}$ has attractor the rectangle $[-1, 1] \times [-\sqrt{2}, \sqrt{2}]$ (one verifies that f_1 and f_2 map it to the top and bottom halves); this rectangle tiles the plane by translation.

Tame twindragon. The entry $(N, \ell) = (2, 1)$ gives $\lambda = (1 + i\sqrt{7})/2$ and the attractor is the *tame twindragon* [1, p. 550], the attractor of $\{f_{1,2}(z) = a(z \pm 1)\}$ with $a = (1 - i\sqrt{7})/4$. One computes $1 - a^2 = (11 + i\sqrt{7})/8$, hence $|1 - a^2| = \sqrt{2}$, and Corollary 3 gives

$$\mathrm{AR}^2 = \frac{\sqrt{2} - \frac{1}{2}}{\sqrt{2} + \frac{1}{2}} = \frac{2\sqrt{2} - 1}{2\sqrt{2} + 1} = \frac{9 - 4\sqrt{2}}{7} \approx 0.478, \qquad \mathrm{AR} \approx 0.691.$$

This tile arises as the numeration digit set for the complex base $\beta = (1 + i\sqrt{7})/2$ (satisfying $\beta^2 - \beta + 2 = 0$ and $|\beta|^2 = 2$) via $a = 1/\beta$;[1] it is marked in Figure 2.

[1]The attractor is the set $\{\sum_{k \geq 1} d_k \beta^{-k} : d_k \in \{-1, +1\}\} \subset \mathbb{C}$.

The connection to metallic ratios in part (iii) is not accidental. The m-th metallic ratio $\mu_m = (m + \sqrt{m^2+4})/2$ is the fundamental unit of $\mathbb{Z}[\mu_m] \subset \mathbb{Q}(\sqrt{m^2+4})$. In the $\ell = 2$ sub-family, $\sqrt{m^2+4} = \sqrt{N+3}$ and the Cayley-type transform (9) maps $(N+1)^2 - 4 = (N-1)(N+3)$ into $\mathbb{Q}(\sqrt{N+3})$. For $N = 2$ and $N = 5$ the aspect ratio is an exact power of the fundamental unit ($1/\varphi$ and $\sqrt{2}-1$ respectively); for other N it remains in $\mathbb{Q}(\sqrt{N+3})$ but is not generally a unit.

Remark 11. Both the aspect ratio and the principal-axis orientation of the $\lambda_{N,\ell}$-family are governed by a single element of the ring of integers: $\lambda^2 - 1 = \ell\lambda - (N+1) \in \mathbb{Z}[\lambda]$. Its norm $(N+1)^2 - \ell^2$ determines AR via (10), and its argument determines ψ via (11). This admits a hierarchical interpretation. The random-iteration representation $Z = \sum_{n\geq 1} a^n \varepsilon_n$ (ε_n i.i.d. from the centred digits) decomposes $E[Z^2]$ as $\sum_{n\geq 1} \sigma^2 a^{2n}$: each level n of the IFS contributes σ^2/N^n to the anisotropy along direction $2n \arg(a) = -2n \arg(\lambda)$. The geometric sum of these rotated contributions points in the direction $\arg\big(a^2/(1-a^2)\big) = -\arg(\lambda^2 - 1)$, which is precisely twice the principal-axis angle. At the boundary $\ell = 0$ the vector $\lambda^2 - 1 = -(N+1)$ is real negative, giving $\psi = -\pi/2$ (vertical major axis); as $\ell \to 2\sqrt{N}$ the vector tends to $2(N-1) > 0$, giving $\psi \to 0$ (horizontal, degenerate).

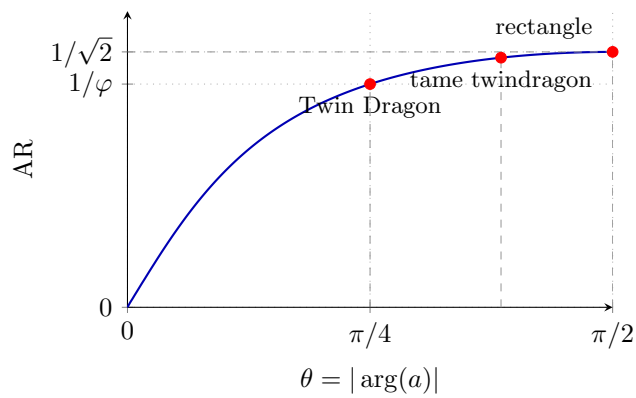


FIGURE 2. Aspect ratio $\mathrm{AR}(\theta)$ of the attractor of $\{a(z \pm 1)\}$ for $|a| = 1/\sqrt{2}$, as a function of $\theta = |\arg(a)|$. The Twin Dragon ($\theta = \pi/4$), the tame twindragon ($\theta = \arctan\sqrt{7}$), and the rectangle attractor ($\theta = \pi/2$) are marked in red.

4. Tiles over $\mathbb{Z}[i]$

In this section we specialise to tiles over $\mathbb{Z}[i]$: the contraction is $a = 1/\lambda$ for a Gaussian integer λ and the translations form a CRS modulo λ. Corollary 3 then applies with $|a|^2 = 1/N$ and κ determined by the choice of CRS.

The dependence on κ is already visible for a single expansion factor. Take $\lambda = 2 + i$ ($N = 5$). The collinear digit set $\{0, 1, 2, 3, 4\}$ gives $\kappa = 1$ and $\mathrm{AR} = 1/\varphi^3$ (row $(5, 4)$ of the table in Section 3). Replacing it by the two-dimensional CRS $\{0,\ 1,\ -i,\ i,\ 1+i\}$ (which is still a complete residue system of $\mathbb{Z}[i]$ modulo $2+i$, since

$i \equiv 3 \pmod{2+i}$) yields $\sigma^2 = 4/5$, $\tau = (-8+6i)/25$, and $\kappa = 1/2$. Substituting into (7):

$$\mathrm{AR}^2 = \frac{\sqrt{5}-1}{\sqrt{5}+1} = \frac{3-\sqrt{5}}{2} = \frac{1}{\varphi^2}, \qquad \mathrm{AR} = \frac{1}{\varphi}.$$

Thus the same golden ratio governs this 5-map tile with $\kappa = 1/2$ as it does the Twin Dragon ($N=2$, $\kappa=1$), while collinear digits for the same λ give the smaller value $1/\varphi^3$ (Figure 3). A natural question arises: for which κ does the aspect ratio equal $1/\mu_m$, the reciprocal of the m-th metallic mean?

Proposition 12. *Let $\lambda = p + qi \in \mathbb{Z}[i] \setminus \mathbb{R}$ with $|\lambda|^2 = N$, and let $m \geq 1$ be a positive integer. The digit anisotropy required for $\mathrm{AR} = 1/\mu_m$ (the reciprocal m-th metallic mean $\mu_m = (m + \sqrt{m^2+4})/2$) is*

$$\kappa_m^* = \frac{m\,|\lambda^2 - 1|}{\sqrt{m^2+4}\,(N-1)}, \tag{12}$$

and $\kappa_m^* \leq 1$ if and only if $mq \leq N-1$. In particular:

(a) *For $m = 1$ (golden ratio), $\kappa_1^* \leq 1$ for every λ, with equality only for $\lambda = 1+i$.*
(b) *The collinear case $\kappa = 1$ corresponds to $m = (N-1)/q$, provided this is an integer; thus the collinear aspect ratio of Corollary 4 equals $1/\mu_{(N-1)/q}$.*
(c) *For $q = 1$ (i.e. $\lambda = p+i$), every metallic ratio $\mu_1, \ldots, \mu_{p^2}$ is accessible.*

Proof. Setting $\mathrm{AR}^2 = 1/\mu_m^2$ in (7) and solving for κ gives $\kappa = |1-a^2| \cdot m/\big((1-|a|^2)\sqrt{m^2+4}\big)$; substituting $a = 1/\lambda$, $|a|^2 = 1/N$, $|1-a^2| = \sqrt{\mathbf{N}(\lambda^2-1)}/N$ yields (12).

The condition $\kappa_m^* \leq 1$ becomes $m^2|\lambda^2-1|^2 \leq (m^2+4)(N-1)^2$. Since $|\lambda^2-1|^2 = (N-1)^2 + 4q^2$, this simplifies to $4m^2q^2 \leq 4(N-1)^2$, i.e. $mq \leq N-1$.

(a): $q \leq N-1$ holds since $N = p^2+q^2 \geq q+1$ for $p \geq 1$. Equality $q = N-1$ gives $p^2 = 1 - q(q-1)$; for $q = 1$ this yields $p = 1$, $\lambda = 1+i$; for $q \geq 2$ one has $p^2 < 0$, which is impossible.

(b): $\kappa_m^* = 1$ iff $m^2q^2 = (N-1)^2$, so $m = (N-1)/q$.

(c): $q = 1$ gives $mq = m \leq N-1 = p^2$. □

Corollary 13. *Up to associates, $\kappa_1^* = 1/2$ if and only if $\lambda \in \{2+i,\ 2+3i,\ 1+4i\}$ ($N = 5, 13, 17$). More generally, $\kappa_1^* = 1/m$ for an integer $m \geq 1$ only for $m = 1$ ($\lambda = 1+i$) and $m = 2$ (the three primes above).*

Proof. Write $\lambda = p+qi$, $N = p^2+q^2$. Setting $\kappa_1^* = 1/m$ in $|\lambda^2-1|^2 = 5(N-1)^2/m^2$ and using $|\lambda^2-1|^2 = (N-1)^2 + 4q^2$ gives

$$4q^2 = (N-1)^2\big(\tfrac{5}{m^2} - 1\big).$$

For $m = 1$ this is $4q^2 = 4(N-1)^2$, so $q = N-1 = 1$, $N = 2$: only $\lambda = 1+i$. For $m = 2$: $4q^2 = \frac{1}{4}(N-1)^2$, so $N = 4q+1$, $p^2 = 4q+1-q^2$, and the positivity condition $p^2 \geq 1$ becomes $p^2 + (q-2)^2 = 5$, i.e. a representation of 5 as a sum of two squares. The three solutions with $q \geq 1$, $p \geq 1$ are $(p,q) = (2,1), (2,3), (1,4)$. For $m \geq 3$: $5/m^2 - 1 < 0$, so $4q^2 < 0$ — impossible. □

Theorem 14. *For every integer $m \geq 1$:*

(a) *There exists a self-similar plane-filling tile in $\mathbb{R}^2$ whose aspect ratio equals $1/\mu_m$, the reciprocal of the m-th metallic mean.*
(b) *Any such tile over $\mathbb{Z}[i]$ requires at least $N_{\min}(m) = \lceil \sqrt{m}\,\rceil^2 + 1$ maps.*

(c) *When m is a perfect square, $N_{\min}(m) = m + 1$, achieved by the collinear CRS for $\lambda = \sqrt{m} + i$.*

Proof. (a): Let $\lambda = 1 + mi \in \mathbb{Z}[i]$. Then $N = |\lambda|^2 = m^2 + 1$ and $\mathrm{Im}(\lambda) = m$. The standard digits $\{0, 1, \ldots, m^2\}$ form a CRS modulo λ (§3); the induced N-map IFS satisfies the open set condition [1], so its attractor is a plane-filling tile. The collinear digit set has $\kappa = 1$, and Proposition 12(b) gives $m = (N-1)/\mathrm{Im}(\lambda) = m^2/m$, hence $\mathrm{AR} = 1/\mu_m$.

(b): For any $\lambda = p+qi$ with $mq \leq N-1$ (Proposition 12), we have $N = p^2+q^2 \geq q(m-q) + 1 + q^2 = qm + 1$, which is minimised at $q = 1$, giving $N \geq \lceil\sqrt{m}\,\rceil^2 + 1$.

(c): When $m = k^2$, the choice $\lambda = k + i$ yields $N = k^2 + 1 = m + 1$ and the collinear CRS has $\kappa = 1$ with $(N-1)/q = m$. □

Remark 15. Not every integer $N \geq 2$ is the norm of a Gaussian integer with non-zero imaginary part. By the sum-of-two-squares theorem, $N = p^2 + q^2$ with $q \neq 0$ if and only if every prime factor of N congruent to 3 (mod 4) appears to an even power. The first admissible values are $N = 2, 4, 5, 8, 9, 10, 13, \ldots$; in particular $N = 3, 6, 7, 11, 14, 15$ are not representable.

Remark 16. For *collinear* digits ($\kappa = 1$) the condition $m = (N-1)/q$ (Proposition 12(b)) requires $p^2 = q(m-q) + 1$ to be a perfect square. Taking $q = m$ gives the universal upper bound $N_{\mathrm{col}}(m) \leq m^2 + 1$, but smaller q may yield a smaller N. For $m = k^2$ a perfect square, $q = 1$ gives $N_{\mathrm{col}}(k^2) = k^2 + 1 = N_{\min}(k^2)$, so the collinear and non-collinear minima coincide. In contrast, for non-square m a gap opens: e.g. $N_{\min}(5) = 10$ while $N_{\mathrm{col}}(5) = 26$, and $N_{\min}(7) = 10$ while $N_{\mathrm{col}}(7) = 50$. In general, $N_{\mathrm{col}}(m)$ equals the minimum of $q^2+q(m-q)+1$ over those $q \in \{1, \ldots, m\}$ for which $q(m-q) + 1$ is a perfect square — a Diophantine condition for which no closed-form solution is known for general m.

Remark 17. Item (b) of Proposition 12 recovers every row of Table 1 as a special case: $\lambda = 1+i$ gives $m = 1$ and $\mathrm{AR} = 1/\mu_1 = 1/\varphi$; $\lambda = 2+i$ gives $m = 4$ and $\mathrm{AR} = 1/\mu_4 = 1/(2+\sqrt{5}) = \sqrt{5} - 2 = 1/\varphi^3$; $\lambda = 1+2i$ gives $m = 2$ and $\mathrm{AR} = 1/\mu_2 = \sqrt{2} - 1$. Thus the collinear aspect ratios of Section 2 and the golden-ratio tiles of Table 2 are part of a single metallic-ratio spectrum parametrised by κ. Table 2 lists CRS digit sets achieving κ_1^* (i.e. $\mathrm{AR} = 1/\varphi$) for small Gaussian primes λ.

TABLE 2. Complete residue systems $D \subset \mathbb{Z}[i]$ (modulo λ, centred to $\sum t_k = 0$) achieving $\mathrm{AR} = 1/\varphi$.

λ	N	κ^*	digit set D
$1+i$	2	1	$\{0, \pm1\}$
$2+i$	5	$1/2$	$\{0, 1, -i, i, 1+i\}$
$1+2i$	5	$\sqrt{2/5}$	$\{-1-2i, -1, i, 1, 1+i\}$
$1+3i$	10	$\sqrt{13/45}$	$\{-2, -2+2i, -1-i, -1, -1+i, -i, 0, i, 1-2i, 1\}$
$2+3i$	13	$1/2$	$\{-2-3i, -2, -1\pm i, -1, \pm2i, \pm i, 1, 1+i, 2, 3+2i\}$

Rows 3 and 4 have irrational κ^* ($\sqrt{2/5}$ and $\sqrt{13/45}$), yet Gaussian-integer digits realise these values exactly. Corollary 13 explains why rows 2 and 5 share $\kappa^* = 1/2$: only three Gaussian primes ($2+i$, $2+3i$, $1+4i$) attain this value, and the proof reduces to the unique representation $5 = 1^2 + 2^2$. A CRS for the remaining prime

$\lambda = 1+4i$ ($N=17$) achieving $\kappa=1/2$ also exists; its 17 digits are omitted from the table for space.

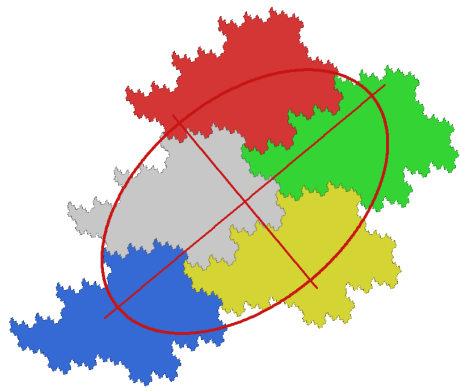


FIGURE 3. The 5-map tile with $\lambda = 2 + i$ and digit set $\{0, 1, -i, i, 1+i\}$ ($\kappa = 1/2$). Its 1.5-sigma principal-axis ellipse (red) confirms AR $= 1/\varphi \approx 0.618$, the same value as the Twin Dragon (Figure 1a).

Lemma 18. *For $\lambda \in \mathbb{C} \setminus \mathbb{R}$ with $|\lambda|^2 = N \geq 2$,*

$$|\lambda^2 - 1|^2 = (N-1)^2 + 4\,(\operatorname{Im}\lambda)^2. \tag{13}$$

Consequently, the digit anisotropy (12) *satisfies $\kappa_m^* \leq 1$ if and only if $m \leq (N-1)/|\operatorname{Im}\lambda|$. The collinear case $\kappa = 1$ corresponds to $m = (N-1)/|\operatorname{Im}\lambda|$, which is a positive integer only when $|\operatorname{Im}\lambda|$ divides $N-1$.*

Proof. Write $\lambda = x + iy$, so $N = x^2 + y^2$ and $\lambda^2 - 1 = (x^2-y^2-1) + 2ixy$. Then

$$|\lambda^2 - 1|^2 = (x^2-y^2-1)^2 + 4x^2y^2 = (x^2+y^2-1)^2 + 4y^2 = (N-1)^2 + 4y^2,$$

since, by direct expansion, $(x^2-y^2-1)^2 + 4x^2y^2 - (x^2+y^2-1)^2 = 4y^2$. Substituting into (12): $\kappa_m^* \leq 1$ iff $m^2\big((N-1)^2 + 4y^2\big) \leq (m^2+4)(N-1)^2$, i.e. $4m^2y^2 \leq 4(N-1)^2$, so $m \leq (N-1)/|y|$. □

Corollary 19. *Among all imaginary quadratic integer rings, $\mathbb{Z}[i]$ is the unique one where collinear self-similar tiles can have metallic-ratio aspect ratios.*

Proof. For $\lambda = p + q\sqrt{-k} \in \mathbb{Z}[\sqrt{-k}]$ ($k \geq 2$, square-free, $q \neq 0$), $\operatorname{Im}\lambda = q\sqrt{k} \notin \mathbb{Q}$, so $(N-1)/\operatorname{Im}\lambda$ is irrational. The same holds for the full ring of integers of $\mathbb{Q}(\sqrt{-3})$: if $\lambda = a + b\omega \in \mathbb{Z}[\omega]$ with $\omega = e^{2\pi i/3}$ and $b \neq 0$, then $\operatorname{Im}\lambda = b\sqrt{3}/2 \notin \mathbb{Q}$. In $\mathbb{Z}[i]$, however, $\operatorname{Im}\lambda = q \in \mathbb{Z}$, and taking $q = 1$ gives $m = (N-1)/1 = p^2$, always a positive integer. □

5. The Twin Dragon

The Twin Dragon $A \subset \mathbb{C}$ is the attractor of the IFS [5]

$$A = f_1(A) \cup f_2(A), \qquad f_{1,2}(z) = \frac{1-i}{2}(z \pm 1). \tag{14}$$

Both maps have the same contraction ratio $r = |(1-i)/2| = 1/\sqrt{2}$, so the similarity dimension is $\alpha = -\log 2/\log r = 2$. The open set condition holds [1, Corollary 3]: the Twin Dragon is a digit tile for $\mathbb{Z}[i]/(1+i)\mathbb{Z}[i]$, so its translates by $\mathbb{Z}[i]$ tile $\mathbb{C}$ with the interiors pairwise disjoint. By [7, §5] the self-similar measure therefore coincides with $\mathcal{H}^\alpha|_A$. Since $\alpha = 2$ equals the ambient dimension, $\mathcal{H}^2$ is Lebesgue area, and the covariance matrix M describes the actual geometric shape (Figure 1a).

Substituting $a = (1-i)/2$ into Theorem 2 (with $N = 2$, $t_k = \pm 1$, so $\sigma^2 = 1$): $|a|^2 = \frac{1}{2}$, $a^2 = -\frac{i}{2}$, $1 - a^2 = \frac{2+i}{2}$, so

$$E[Z^2] = \frac{a^2}{1-a^2} = \frac{-i}{2+i} = \frac{-1-2i}{5}, \qquad E[|Z|^2] = \frac{|a|^2}{1-|a|^2} = 1.$$

The covariance matrix follows via $M_{11} = \frac{1}{2}(E[|Z|^2] + \operatorname{Re} E[Z^2])$, $M_{22} = \frac{1}{2}(E[|Z|^2] - \operatorname{Re} E[Z^2])$, $M_{12} = \frac{1}{2}\operatorname{Im} E[Z^2]$:

$$M = \frac{1}{5}\begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix}. \tag{15}$$

The matrix M has $\operatorname{tr} M = 1$ and $\det M = \frac{1}{5}$, so its eigenvalues are

$$I_{1,2} = \frac{1 \mp \frac{1}{\sqrt{5}}}{2}. \tag{16}$$

The geometric aspect ratio — the ratio of standard deviations along the principal axes of M — is

$$\sqrt{\frac{I_1}{I_2}} = \sqrt{\frac{\sqrt{5}-1}{\sqrt{5}+1}} = \sqrt{\frac{3-\sqrt{5}}{2}} = \frac{1}{\varphi}, \tag{17}$$

where $\varphi = (1+\sqrt{5})/2$ is the golden ratio.

Corollary 20. *The attractor of the IFS* $\{f_{1,2}(z) = \frac{1-i}{2}(z \pm 1)\}$ *has covariance matrix* $M = \frac{1}{5}\left(\begin{smallmatrix} 2 & -1 \\ -1 & 3 \end{smallmatrix}\right)$*, geometric aspect ratio* $1/\varphi = (\sqrt{5}-1)/2 \approx 0.618$*, and major-axis direction* $(1, -\varphi)$ *(angle* $-\arctan\varphi \approx -58.3°$ *with the horizontal), where* $\varphi = (1+\sqrt{5})/2$ *is the golden ratio.*

Proof. This is the case $\lambda = 1 + i$ of Corollary 4 ($\mathbf{N}(\lambda^2 - 1) = 5$, $d = 5$), with the orientation from Proposition 10(ii). □

Remark 21. The Twin Dragon is the $(N, \ell) = (2, 2)$ instance of Proposition 10, giving $\tan\psi = -\varphi$ (i.e. $\psi \approx -58.3°$). Thus the same constant φ governs both the aspect ratio ($\mathrm{AR} = 1/\varphi$) and the orientation ($\tan\psi = -\varphi$).

Remark 22. The appearance of φ is unexpected: the Twin Dragon is defined by the Gaussian integer $1 + i$, with no pentagon, decagon, or Fibonacci recurrence in the construction. The number 5 — and hence φ — can be traced to a single Gaussian prime. By the ergodic theorem for random iteration [3], μ is the distribution of the random series $Z = \sum_{k\geq 1} \varepsilon_k a^k$ with $a = (1-i)/2$ and $\varepsilon_k \overset{\text{iid}}{\sim} \{\pm 1\}$. Since $1 - a^2 = (2+i)/2$ and $\kappa = 1$ (real translations), the ellipticity is $\rho = (1-|a|^2)\kappa/|1-a^2| = 1/\sqrt{5}$,

where $5 = \mathbf{N}(2+i)$ is the norm of the Gaussian prime $2+i$. Thus φ enters via the Gaussian factorisation $5 = (2+i)(2-i)$ in $\mathbb{Z}[i]$, and the identity $\mathrm{AR}^2 = (1-\rho)/(1+\rho)$ from (7) immediately gives $1/\varphi^2$ (using $\varphi^2 = \varphi + 1$).

Further directions. An explicit closed-form construction of a CRS achieving $\kappa = \kappa_m^*$ for all N would be desirable; this amounts to a system of quadratic Diophantine equations in the digit shifts $c_k \in \mathbb{Z}[i]$ (where $d_k = k + c_k\lambda$), which does not appear to simplify into a universal formula. More broadly, for a fixed expansion factor λ, characterising the set of all achievable anisotropy values κ — and in particular the CRS that minimise or maximise κ — is a natural combinatorial-geometric problem.

The equal-contraction assumption ($a_k \equiv a$) is essential to our closed-form formulas. For an IFS $\{a_k z + t_k\}$ with distinct contraction ratios the fixed-point equation for the second moment remains linear and is solvable in closed form for small N; extending the aspect-ratio classification to this setting is a natural next step. In a different direction, higher-order moments ($E[Z^3]$, $E[Z^4]$, $\dots$) satisfy analogous fixed-point equations and yield shape descriptors beyond the aspect ratio (skewness, kurtosis); their systematic study for self-similar measures remains open.

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